

4.1 J-mennös regressio.

$$z = ax + by + c$$

$$\left. \begin{array}{l} 1. (0, 0, 1) \Rightarrow 1 = c \\ 2. (1, 0, 1) \Rightarrow 1 = a + c \\ 3. (0, 1, 1) \Rightarrow 1 = b + c \\ 4. (1, 1, 2) \Rightarrow 2 = a + b + c \\ 5. (1, -1, 0) \Rightarrow 0 = a - b + c \\ 6. (-1, 1, 1) \Rightarrow 1 = -a + b + c \end{array} \right\} \begin{array}{c} a \quad b \quad c \\ \left(\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ -1 & 1 & 1 & 1 \end{array} \right) = (A|b) \end{array}$$

$$\text{Eli: } \langle A\hat{x}, A\hat{x} - b \rangle = 0, \text{ missi: } A\hat{x} - b \perp \text{Im } A$$

$$\langle A^T(A\hat{x} - b) \rangle = 0$$

$$A^T(A\hat{x} - b) = 0 \Rightarrow A^T A \hat{x} = A^T b$$

$$\text{istän tölly rumien } A^T A \hat{x} = A^T b$$

$$A^T A = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix}$$

$$A^T b = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 6 \end{pmatrix}$$

$$\text{dastävinnä} \quad \left(\begin{array}{ccc|c} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 2 \\ 2 & 2 & 6 & 6 \end{array} \right) \sim \left(\begin{array}{ccc|c} 4 & 3 & 3 & 3 \\ -1 & 5 & 7 & 7 \\ 4 & -1 & 2 & 2 \end{array} \right) \sim \left(\begin{array}{ccc|c} 4 & 3 & 3 & 3 \\ 0 & 5 & 5 & 7 \\ 0 & -5 & -10 & -2 \end{array} \right) \sim \left(\begin{array}{ccc|c} 4 & 3 & 3 & 3 \\ 0 & 5 & 5 & 7 \\ 0 & 0 & -5 & -3 \end{array} \right)$$

$$\sim \left(\begin{array}{ccc|c} 4 & 3 & 3 & 3 \\ 0 & 5 & 5 & 7 \\ 0 & 0 & -5 & -3 \end{array} \right) \quad c = \frac{2}{5}, \quad b = \frac{4}{5}$$

$$a = 3 - \frac{4}{5} - 3 \cdot \frac{2}{5} = \frac{15 - 4 - 6}{5} = \frac{5}{5} = 1$$

$$(a, b, c) = \left(\frac{5}{5}, \frac{4}{5}, \frac{2}{5} \right)$$

$$\text{Missi jän nuvin } z = \frac{1}{5}x + \frac{4}{5}y + \frac{2}{5} \quad \text{takom, se je}$$

$$\sum_i \left(z_i - \left(\frac{1}{5}x_i + \frac{4}{5}y_i + \frac{2}{5} \right) \right)^2 \quad \text{minimäsi}$$